BEAM DESIGN CALCULATION

Project Name	:	G+2 RB BEAM DESIGN @FLOOR 1
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Beam-1-5.029 m.R1.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 00:54:36
Level Designed	:	5.029 m

Sr.No.	Symbol		Definitions
1	α	=	Angle formed with horizontal by diagonal reinforcement
2	ε1	=	Strain at level considered, Calculated ignoring the stiffening of the concrete in tension zone
3	εm	=	Average steel strain at level considered
4	Acr	=	Distance from the point considered to the surface of the nearest longitudinal bar
5	Adr	=	Area of diagonal reinforcement in coupler beam
6	AstCalc	=	Area of Tension reinforcement required in sqmm
7	Ast comp	=	Ast at compression side
8	AstCrack	=	Area of Tension reinforcement for Crack Width required in sqmm
9	AstPrv	=	Area of reinforcement provided at given section in sqmm
10	Asv	=	Area of shear reinforcement required per meter length in sqmm
11	Asv Reqd	=	Min area shear reinforcement as per clause 41.4.3
12	Asv Total Prv	=	Total area of shear reinforcement provided, including that for torsion
13	Asv Torsion	=	Area of Stirrups over the corner bars to resist shear and torsion as per clause 41.4.3
14	Asv Torsion Prv	=	Area of shear reinforcement resisting torsion (outer legs only)
15	b	=	Width of the Beam in mm
16	bw	=	Width of Web in mm
17	Ceff	=	Effective Cover of Beam in mm
18	Cmin	=	Clear cover to top and bottom reinforcement in mm
19	Cside	=	Clear cover to side reinforcement in mm
20	d	=	Effective depth of Beam in mm
21	d'	=	Clear cover to reinforcement at compression side.
22	D	=	Depth of Beam in mm
23	D'	=	Distance (c/c) between top-most and bottom-most layer
24	Df	=	Thickness of Flange in mm
25	Ec	=	Modulus of elasticity of concrete
26	Es	=	Modulus of elasticity of steel
27	FcPerm	=	Permissible Stress in Concrete required in N/sqmm

28	Fst	=	Stress in steel
29	FstPerm	=	Permissible Stress in Steel required in N/sqmm
30	Legs	=	Number of legs of the shear reinforcement
31	lo	=	Effective Length of Beam in mm
32	Mh	=	Hogging Moment of Resistance in kNm
33	Ms	=	Sagging Moment of Resistance in kNm
34	M_{Tu}	=	Converted Bending Moment due to Torsion at a section in kNm
35	Mu	=	Factored Bending Moment in Limit State at a section in kNm
36	Mud	=	Total Design Moment = Mu + M_{Tu} in kNm
37	MuLim	=	Limiting Moment of Resistence of Singly Reinforced Section
38	NA angle (φ)	=	It is a neutral axis angle corresponding to load angle to find out MCap
39	Pcclc	=	Calculated percentage compression steel
40	Ptclc	=	Calculated percentage tension steel
41	Ptmin	=	Minimum percentage steel as per code
42	PtPrv	=	provided percentage steel
43	R	=	Moment Ratio Mud/Bd^2
44	Reinf	=	Rebar arrangement provided
45	sp	=	Spacing Between bars at outer most layer
46	Spc1 to Spc6	=	Max Stirrup Criteria in mm
47	Stirrup Dia	=	Diameter of shear stirrups in mm
48	SvCalc	=	Stirrup spacing calculated as per Asv in mm
49	SvPrv	=	Stirrup spacing provided in m
50	Tc	=	Permissible Shear stress in concrete at a given section in N/sqmm Calculated as per Tensile reinforcement
51	Tce	=	Earthquake inducted shear stress in N/sqmm
52	Tcep	=	Permissible value of shear stress for checking of coupling action in N/sqmm
53	Tu	=	Factored Torsional Moment in Limit State at a section in kNm
54	Tv	=	Shear stress due to Vud in N/sqmm
55	V^{D+L}	=	Shear force 1.2x(Dead Load + Live Load) in kN
56	Vc	=	Shear capacity of concrete section due to Tc in kN
57	Vu	=	Factored Shear Force in Limit State at a section in kN
58	Vu-A1	=	V^{D+L}Left - 1.4 x (MsLeft + MhRight)/L in kN
59	Vu-A2	=	V^{D+L}Left + 1.4 x (MhLeft + MsRight)/L in kN
60	Vu-B1	=	V^{D+L}Right + 1.4 x (MsLeft + MhRight)/L in kN
61	Vu-B2	=	V^{D+L}Right - 1.4 x (MhLeft + MsRight)/L in kN
62	Vud	=	Design Shear Force (Max of Vus and Vu-sway) in kN
63	Vue	=	Earthquake inducted shear in kN
64	Vu-sway	=	Max of (Vu-A1 and Vu-A2) & (Vu-B1 and Vu-B2) in kN
65	Vus	=	Vud - Vc Design Shear Force for reinforcement, in kN
66	Vut	=	Total Shear Force = Vu + V_{Tu} in kN

67	V_{Tu}	=	Converted Shear Force due to Torsion at a section in kN
68	Wcr	=	Surface Crack Width
69	WcrPerm	=	Permissible Crack Width required in mm
70	X1	=	Smaller c/c dimension of outermost Stirrups
71	Y1	=	Larger c/c dimension of outermost Stirrups
			All 6mm diameter bars are assumed 250 grade with fy = 250 N/sqmm

All Forces are in 'kN', 'kNm', Stress in 'N/sqmm' & Dimension are in 'mm'.

Code References

IS 456

Sr.No.	Item		Clause / Table
1.	Ptmax	:	26.5.1.1
2.	Ptmin	:	26.5.1.1
3.	M_{Tu}	:	41.4.2
4.	V_{Tu}	:	41.3.1
5.	Ptclc	:	38
6.	Tc	:	40.2.1
7.	Tcmax	:	40.2.3
8.	Asv	:	40.4
9.	Min Shear Reinf	:	26.5.1.6
10.	Max Stirrup Spacing	:	26.5.1.5
11.	Shear Reinf - Torsion	:	26.5.1.7
12.	Side Face Reinforcement	:	26.5.1.3
13.	crack width calculation	:	Annex-F
14.	cracking	:	Clause 35.3.2

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Sr.No.	Item		Clause / Table
1.	Ptmax	:	6.2.2
2.	Ptmin	:	6.2.1
3.	Astmin	:	6.2.3 & 6.2.4
4.	Svclc	:	6.3.5
5.	Shear Capacity	:	6.3.4
6.	Coupling Beams	:	10.5

Group	:	**G1**	
Beam No	:	**B1**	
Analysis Reference (Member)	:	3 (5.029 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm

Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	12	19	14	19	12
Critical L/C - RCDC	13	8	15	10	15	8
Mu (kNm)	75.11	48.59	40.82	85.93	36.62	91.47
Tu (kNm)	0.54	0.67	0.17	0	0.17	0.67
M_{Tu} (kNm)	0.91	1.12	0.28	0	0.28	1.12
Mud (kNm)	76.03	49.71	41.1	85.93	36.9	92.6
MuLim (kNm)	102	102	102	102	102	102
R	2.415	1.579	1.305	2.729	1.172	2.941
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.579	0.358	0.292	0.669	0.26	0.734
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	492.42	304.93	248.12	569.69	221.14	624.25
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
				Area of reinforcement required for

D = AstCrack = Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	64.41	53.55	77.1
Tu (kNm)	0	0.66	0.66
V_{Tu} (kN)	0	4.63	4.63
Vut (kN)	64.41	58.18	81.73
Tv (N/sqmm)	0.76	0.68	0.96
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	7.72	1.49	25.04
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	-	204.606	279.167
Asv Torsion Prv (sqmm)	-	365.6	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	140
SvPrv (mm)	275	275	140
Asv Total Prv (sqmm)	365.6	365.6	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

Group	:	**G1**	
Beam No	:	**B2**	
Analysis Reference (Member)	:	4 (5.029 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	19	19	14	12	12
Critical L/C - RCDC	13	15	15	10	8	8
Mu (kNm)	47.92	42.73	60.01	79.36	47.88	111.84
Tu (kNm)	0	0.56	0.56	0.72	0	0
M_{Tu} (kNm)	0	0.94	0.94	1.22	0	0
Mud (kNm)	47.92	43.67	60.95	80.58	47.88	111.84
MuLim (kNm)	102	102	102	102	102	102
R	1.522	1.387	1.936	2.559	1.521	3.552
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.344	0.311	0.449	0.62	0.344	0.908
Pcclc (%) (C)	0	0	0	0	0	0.088
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	292.93	264.9	382.23	527.39	292.69	772.83
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A = Ptmin = Min % area of flexural reinforcement

Maximum of % reinforcement required for

B'	=	Ptclc	=	Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	67.55	50.5	74.05
Tu (kNm)	0.72	0	0
V_{Tu} (kN)	5.03	0	0
Vut (kN)	72.58	50.5	74.05
Tv (N/sqmm)	0.85	0.59	0.87
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	15.89	0	17.36
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	251.952	-	-
Asv Torsion Prv (sqmm)	718.14	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm

Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group	:	**G1**	
Beam No	:	**B3**	
Analysis Reference (Member)	:	140 (5.029 m)	
Beam Length	:	1372	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	15	15	15	15	15	15
Critical L/C - RCDC	11	11	11	11	11	11
Mu (kNm)	-	-	-	36.12	26.2	9
Tu (kNm)	15.62	15.62	15.62	15.62	15.62	15.62
M_{Tu} (kNm)	26.37	26.37	26.37	26.37	26.37	26.37
Mud (kNm)	26.37	26.37	26.37	62.49	52.57	35.37
MuLim (kNm)	102	102	102	102	102	102
R	0.837	0.837	0.837	1.985	1.669	1.123
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.462	0.381	0.248
Pcclc (%) (C)	0.182	0.182	0.182	0	0.001	0.118
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	393.1	324.21	211.38
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	15	15	15
Critical L/C - RCDC	11	11	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	30.22	26.79	23.36
Tu (kNm)	15.62	15.62	15.62
V_{Tu} (kN)	108.68	108.68	108.68
Vut (kN)	138.9	135.47	132.04
Tv (N/sqmm)	1.63	1.59	1.55
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	82.21	78.78	75.35
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	919.844	908.983	898.121
Asv Torsion Prv (sqmm)	957.52	914	914
Asv Reqd (sqmm/m)	615.43	589.75	564.06
SvCalc (mm)	105	110	110
SvPrv (mm)	105	110	110
Asv Total Prv (sqmm)	957.52	914	914

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm
Side Face Reinforcement not required.		

Group	:	**G2**	
Beam No	:	**B4**	
Analysis Reference (Member)	:	23 (5.029 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	12	12	19	19	19	12
Critical L/C - RCDC	8	8	15	15	15	8
Mu (kNm)	88.33	45.81	53.51	85.14	42.68	85.45
Tu (kNm)	0.3	0.3	0.26	0.26	0.26	0.3
M_{Tu} (kNm)	0.5	0.5	0.43	0.43	0.43	0.5
Mud (kNm)	88.84	46.31	53.94	85.57	43.11	85.95
MuLim (kNm)	102	102	102	102	102	102
R	2.821	1.471	1.713	2.718	1.369	2.73
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.697	0.332	0.392	0.666	0.307	0.67

Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	593.19	282.28	333.61	566.83	261.25	569.83
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	50.77	50.18	61.96
Tu (kNm)	0.24	0.3	0.3
V_{Tu} (kN)	1.7	2.07	2.07
Vut (kN)	52.47	52.25	64.03
Tv (N/sqmm)	0.62	0.61	0.75
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	7.34
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	173.616	174.544	211.824
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G2**	
Beam No	:	**B5**	
Analysis Reference (Member)	:	24 (5.029 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	19	19	14	12	12
Critical L/C - RCDC	13	15	15	10	8	8
Mu (kNm)	61.53	40.88	67.07	71.34	60.58	117.79
Tu (kNm)	0.24	0.25	0.25	0.25	0.23	0.23

M_{Tu} (kNm)	0.4	0.42	0.42	0.43	0.4	0.4
Mud (kNm)	61.93	41.3	67.5	71.77	60.98	118.19
MuLim (kNm)	102	102	102	102	102	102
R	1.967	1.312	2.144	2.279	1.937	3.754
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.457	0.293	0.504	0.541	0.449	0.958
Pcclc (%) (C)	0	0	0	0	0	0.148
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	389.13	249.46	429.06	460.46	382.4	815.6
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	49.72	50.97	62.75
Tu (kNm)	0.25	0.23	0.23
V_{Tu} (kN)	1.77	1.63	1.63
Vut (kN)	51.48	52.6	64.38
Tv (N/sqmm)	0.6	0.62	0.76
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	7.69
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	170.761	173.695	210.975
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81

SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G2**	
Beam No	:	**B6**	
Analysis Reference (Member)	:	145 (5.029 m)	
Beam Length	:	1372	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	13	13	13	8	8	13
Critical L/C - RCDC	9	9	9	4	4	9
Mu (kNm)	-	-	3.22	61.12	44.49	11.19
Tu (kNm)	13.68	13.68	13.68	11.57	11.57	13.68
M_{Tu} (kNm)	23.09	23.09	23.09	19.53	19.53	23.09
Mud (kNm)	23.09	23.09	26.32	80.65	64.02	34.28
MuLim (kNm)	102	102	102	102	102	102
R	0.733	0.733	0.836	2.561	2.033	1.089
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.62	0.475	0.24
Pcclc (%) (C)	0.159	0.159	0.136	0	0	0.08
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	527.96	404.01	204.52
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	5	5	5
Critical L/C - RCDC	1	1	1
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	59.14	58.31	55.73
Tu (kNm)	2.22	2.22	2.22
V_{Tu} (kN)	15.43	15.43	15.43
Vut (kN)	74.57	73.74	71.16
Tv (N/sqmm)	0.88	0.87	0.84
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69

Vus=Vut-Vc (kN)	17.88	17.05	14.47
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	304.207	301.588	293.415
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G3**	
Beam No	:	**B7**	
Analysis Reference (Member)	:	43 (5.029 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover	:	30	mm

(Cmin)			
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	12	12	19	19	19	12
Critical L/C - RCDC	8	8	15	15	15	8
Mu (kNm)	93.54	46.1	61.65	89.55	43.95	94.99
Tu (kNm)	0	0	0.38	0.38	0.38	0
M_{Tu} (kNm)	0	0	0.64	0.64	0.64	0
Mud (kNm)	93.54	46.1	62.29	90.19	44.59	94.99
MuLim (kNm)	102	102	102	102	102	102
R	2.971	1.464	1.978	2.864	1.416	3.017
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.743	0.33	0.46	0.71	0.318	0.757
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	1.107	1.107	1.107	1.107	1.107	1.107
AstCalc (sqmm)	632.18	280.88	391.66	604.28	270.94	644.43
AstPrv (sqmm)	942.48	942.48	942.48	942.48	942.48	942.48
Reinforcement Provided	3-T20	3-T20	3-T20	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	7
Critical L/C - RCDC	10	8	3
PtPrv (%)	1.107	1.107	1.107

Vu (kN)	53.83	53.94	66.1
Tu (kNm)	0.45	0	0
V_{Tu} (kN)	3.14	0	0
Vut (kN)	56.97	53.94	66.1
Tv (N/sqmm)	0.67	0.63	0.78
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0.28	0	9.41
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	194.191	-	-
Asv Torsion Prv (sqmm)	718.14	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

Group : **G3**

Beam No : **B8**

Analysis Reference (Member) : 44 (5.029 m)

Beam Length : 3924 mm

Breadth (B) : 230 mm

Depth (D) : 430 mm

Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	19	19	14	12	12
Critical L/C - RCDC	13	15	15	10	8	8
Mu (kNm)	71.11	43.17	72.85	79.07	60.72	121.34
Tu (kNm)	0.39	0.14	0.14	0	0.47	0.47
M_{Tu} (kNm)	0.65	0.23	0.23	0	0.8	0.8
Mud (kNm)	71.77	43.4	73.09	79.07	61.52	122.13
MuLim (kNm)	102	102	102	102	102	102
R	2.279	1.378	2.321	2.511	1.954	3.879
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.541	0.309	0.553	0.606	0.454	0.99
Pcclc (%) (C)	0	0	0	0	0	0.185
PtPrv (%)	1.107	1.107	1.107	1.107	1.107	1.107
AstCalc (sqmm)	460.46	263.13	470.27	515.69	386.25	842.19
AstPrv (sqmm)	942.48	942.48	942.48	942.48	942.48	942.48
Reinforcement Provided	3-T20	3-T20	3-T20	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	53.3	54.44	66.21
Tu (kNm)	0	0.47	0.47
V_{Tu} (kN)	0	3.28	3.28
Vut (kN)	53.3	57.71	69.49
Tv (N/sqmm)	0.63	0.68	0.82
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	1.02	12.8
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	-	197.167	234.447
Asv Torsion Prv (sqmm)	-	365.6	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	140
SvPrv (mm)	275	275	140
Asv Total Prv (sqmm)	365.6	365.6	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

Group : **G3**

Beam No	:	**B9**	
Analysis Reference (Member)	:	150 (5.029 m)	
Beam Length	:	1372	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	13	13	13	5	8	13
Critical L/C - RCDC	9	9	9	1	4	9
Mu (kNm)	-	-	-	63.82	35.27	11.12
Tu (kNm)	15.25	15.25	15.25	4	13.18	15.25
M_{Tu} (kNm)	25.73	25.73	25.73	6.75	22.26	25.73
Mud (kNm)	25.73	25.73	25.73	70.56	57.53	36.85
MuLim (kNm)	102	102	102	102	102	102
R	0.817	0.817	0.817	2.241	1.827	1.17
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.531	0.421	0.259
Pcclc (%) (C)	0.178	0.178	0.178	0	0	0.099
PtPrv (%)	1.107	1.107	1.107	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	451.53	358.3	220.81
AstPrv (sqmm)	942.48	942.48	942.48	942.48	942.48	942.48
Reinforcement Provided	3-T20	3-T20	3-T20	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A = Ptmin = Min % area of flexural reinforcement

Maximum of % reinforcement required for

B'	=	Ptclc	=	Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	5	5	5
Critical L/C - RCDC	1	1	1
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	48.78	47.96	45.37
Tu (kNm)	4	4	4
V_{Tu} (kN)	27.81	27.81	27.81
Vut (kN)	76.59	75.76	73.18
Tv (N/sqmm)	0.9	0.89	0.86
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	19.9	19.07	16.49
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	365.315	362.695	354.523
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm

Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group	:	**G4**	
Beam No	:	**B10**	
Analysis Reference (Member)	:	63 (5.029 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	12	19	14	19	12
Critical L/C - RCDC	13	8	15	10	15	8
Mu (kNm)	88.46	51.91	56.97	97.91	41.82	108.9
Tu (kNm)	0	0	0.42	0.53	0.42	0
M_{Tu} (kNm)	0	0	0.71	0.9	0.71	0
Mud (kNm)	88.46	51.91	57.68	98.81	42.53	108.9
MuLim (kNm)	102	102	102	102	102	102
R	2.81	1.649	1.832	3.138	1.351	3.458
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.694	0.376	0.422	0.796	0.303	0.885
Pcclc (%) (C)	0	0	0	0	0	0.06
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	590.17	319.76	359.35	677.26	257.46	752.97
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	71.5	61.48	85.04
Tu (kNm)	0.53	0	0
V_{Tu} (kN)	3.7	0	0
Vut (kN)	75.2	61.48	85.04
Tv (N/sqmm)	0.88	0.72	1
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	18.51	4.79	28.35
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	254.393	-	-
Asv Torsion Prv (sqmm)	718.14	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group : **G4**
Beam No : **B11**
Analysis Reference (Member) : 64 (5.029 m)
Beam Length : 3924 mm
Breadth (B) : 230 mm
Depth (D) : 430 mm
Effective Depth (d) : 370 mm
Design Code : IS 456
Beam Type : Regular Beam
Grade Of Concrete (Fck) : M25 N/sqmm
Grade Of Steel (Main) : Fe550 N/sqmm
Grade Of Steel (Shear) : Fe550 N/sqmm
Top/Bottom Clear Cover (Cmin) : 30 mm
Side Clear Cover : 30 mm
Es : 2x10^5 N/sqmm
Ptnominal (%) (Bn) : 0.2 %

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	19	19	14	12	12
Critical L/C - RCDC	13	15	15	10	8	8
Mu (kNm)	64.64	43.95	66.24	92.19	53.95	125.84
Tu (kNm)	0.15	0	0	0.13	0.22	0.22
M_{Tu} (kNm)	0.26	0	0	0.22	0.38	0.38
Mud (kNm)	64.9	43.95	66.24	92.4	54.33	126.22
MuLim (kNm)	102	102	102	102	102	102
R	2.061	1.396	2.104	2.935	1.725	4.009
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.482	0.313	0.494	0.732	0.395	1.022

Pcclc (%) (C)	0	0	0	0	0	0.224
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	410.33	266.73	419.98	622.65	336.25	869.76
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	71.92	58.58	82.13
Tu (kNm)	0.13	0.22	0.22
V_{Tu} (kN)	0.9	1.56	1.56
Vut (kN)	72.82	60.13	83.69
Tv (N/sqmm)	0.86	0.71	0.98
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	16.13	3.44	27
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	234.459	197.214	271.775
Asv Torsion Prv (sqmm)	718.14	365.6	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	140
SvPrv (mm)	140	275	140
Asv Total Prv (sqmm)	718.14	365.6	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G4**	
Beam No	:	**B12**	
Analysis Reference (Member)	:	155 (5.029 m)	
Beam Length	:	1372	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	15	15	13	8	13	13
Critical L/C - RCDC	11	11	9	4	9	9
Mu (kNm)	-	-	8.3	62.46	41.68	8.36
Tu (kNm)	8.1	8.1	7.74	6.12	7.74	7.74

M_{Tu} (kNm)	13.67	13.67	13.06	10.34	13.06	13.06
Mud (kNm)	13.67	13.67	21.36	72.79	54.74	21.42
MuLim (kNm)	102	102	102	102	102	102
R	0.434	0.434	0.678	2.312	1.739	0.68
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.55	0.398	0.2
Pcclc (%) (C)	0.093	0.093	0.032	0	0	0.031
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	468.09	339.07	170.2
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	8	8	13
Critical L/C - RCDC	4	4	9
PtPrv (%)	1.107	1.107	0.709
Vu (kN)	53.86	53.08	48.59
Tu (kNm)	6.12	6.12	7.74
V_{Tu} (kN)	42.61	42.61	53.81
Vut (kN)	96.46	95.69	102.4
Tv (N/sqmm)	1.13	1.12	1.2
Tc (N/sqmm)	0.67	0.67	0.56
Vc (kN)	56.69	56.69	47.85
Vus=Vut-Vc (kN)	39.77	39	54.55
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	493.579	491.134	561.908
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	297.75	291.97	408.38

SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G5**	
Beam No	:	**B13**	
Analysis Reference (Member)	:	136 (5.029 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	7	5	5	14	5	7
Critical L/C - RCDC	3	1	1	10	1	3
Mu (kNm)	15.56	17.19	9.27	10.15	-	10.51
Tu (kNm)	2.33	2.75	2.75	1.41	2.75	2.33
M_{Tu} (kNm)	3.94	4.64	4.64	2.38	4.64	3.94
Mud (kNm)	19.5	21.83	13.91	12.53	4.64	14.44
MuLim (kNm)	102	102	102	102	102	102
R	0.619	0.693	0.442	0.398	0.147	0.459
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2
Pcclc (%) (C)	0	0	0	0	0.031	0
PtPrv (%)	0.266	0.266	0.266	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
AstPrv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement Provided	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	5	7	5
Critical L/C - RCDC	1	3	1
PtPrv (%)	0.399	0.266	0.399
Vu (kN)	19.55	7.5	23.62
Tu (kNm)	2.75	2.33	2.75
V_{Tu} (kN)	19.12	16.22	19.12
Vut (kN)	38.68	23.71	42.74
Tv (N/sqmm)	0.45	0.28	0.5
Tc (N/sqmm)	0.45	0.37	0.45
Vc (kN)	37.88	31.88	37.88

Vus=Vut-Vc (kN)	0.8	0	4.86
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	195.113	137.355	207.685
Asv Torsion Prv (sqmm)	718.14	365.6	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	140
SvPrv (mm)	140	275	140
Asv Total Prv (sqmm)	718.14	365.6	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

Group	:	**G5**	
Beam No	:	**B14**	
Analysis Reference (Member)	:	137 (5.029 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover	:	30	mm

(Cmin)

Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	8	9	14	9	17	12
Critical L/C - RCDC	4	5	10	5	13	8
Mu (kNm)	6.02	12.29	10.43	9.9	12.73	30.92
Tu (kNm)	2.08	2.17	2.05	2.17	0.38	0
M_{Tu} (kNm)	3.52	3.66	3.47	3.66	0.64	0
Mud (kNm)	9.53	15.96	13.9	13.56	13.37	30.92
MuLim (kNm)	102	102	102	102	102	102
R	0.303	0.507	0.441	0.431	0.425	0.982
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.216
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.266	0.266	0.266	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	183.4
AstPrv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement Provided	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	9	7	5
Critical L/C - RCDC	5	3	1
PtPrv (%)	0.399	0.399	0.399

Vu (kN)	19.23	12.55	24.18
Tu (kNm)	2.17	0.56	1.7
V_{Tu} (kN)	15.1	3.87	11.85
Vut (kN)	34.33	16.42	36.03
Tv (N/sqmm)	0.4	0.19	0.42
Tc (N/sqmm)	0.45	0.45	0.45
Vc (kN)	37.88	37.88	37.88
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	165.786	66.079	158.252
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G5**	
Beam No	:	**B15**	
Analysis Reference (Member)	:	158 (5.029 m)	
Beam Length	:	1372	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm

Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	7	7	7	12	12	7
Critical L/C - RCDC	3	3	3	8	8	3
Mu (kNm)	-	-	-	22.09	17.41	10.59
Tu (kNm)	10.76	10.76	10.76	10.45	10.45	10.76
M_{Tu} (kNm)	18.16	18.16	18.16	17.64	17.64	18.16
Mud (kNm)	18.16	18.16	18.16	39.73	35.05	28.75
MuLim (kNm)	102	102	102	102	102	102
R	0.577	0.577	0.577	1.262	1.113	0.913
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.281	0.246	0.2
Pcclc (%) (C)	0.124	0.124	0.124	0	0.001	0.051
PtPrv (%)	0.266	0.266	0.266	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	239.26	209.38	170.2
AstPrv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement Provided	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	12	12	12

Critical L/C - RCDC	8	8	8
PtPrv (%)	0.399	0.399	0.399
Vu (kN)	14.92	11.49	8.06
Tu (kNm)	10.45	10.45	10.45
V_{Tu} (kN)	72.69	72.69	72.69
Vut (kN)	87.61	84.18	80.75
Tv (N/sqmm)	1.03	0.99	0.95
Tc (N/sqmm)	0.45	0.45	0.45
Vc (kN)	37.88	37.88	37.88
Vus=Vut-Vc (kN)	49.73	46.3	42.87
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	557.839	547.22	536.601
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	372.24	346.55	320.86
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	: **G6**	
Beam No	: **B16**	
Analysis Reference (Member)	: 123 (5.029 m)	
Beam Length	: 2134	mm

Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	14	14	14	14	9	9
Critical L/C - RCDC	10	10	10	10	5	5
Mu (kNm)	-	-	-	11.22	41.3	59.13
Tu (kNm)	10.15	10.15	10.15	10.15	8.84	8.84
M_{Tu} (kNm)	17.13	17.13	17.13	17.13	14.92	14.92
Mud (kNm)	17.13	17.13	17.13	28.36	56.22	74.06
MuLim (kNm)	102	102	102	102	102	102
R	0.544	0.544	0.544	0.901	1.786	2.352
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.41	0.561
Pcclc (%) (C)	0.117	0.117	0.117	0.04	0	0
PtPrv (%)	0.399	0.399	0.399	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	170.2	349.26	477.53
AstPrv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement Provided	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required

D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	9	9	9
Critical L/C - RCDC	5	5	5
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	23.36	30.13	35.2
Tu (kNm)	8.84	8.84	8.84
V_{Tu} (kN)	61.49	61.49	61.49
Vut (kN)	84.86	91.63	96.7
Tv (N/sqmm)	1	1.08	1.14
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	28.17	34.94	40.01
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	540.299	561.723	577.773
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	261.55	299.51
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

Group	:	**G6**	
Beam No	:	**B17**	
Analysis Reference (Member)	:	108 (5.029 m)	
Beam Length	:	4496	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	13	13	15
Critical L/C - RCDC	16	9	14	9	9	11
Mu (kNm)	38.89	33.63	43.42	104.14	35.47	80.8
Tu (kNm)	0.46	0.25	0	0.25	0.25	0.63
M_{Tu} (kNm)	0.77	0.42	0	0.42	0.42	1.07
Mud (kNm)	39.66	34.05	43.42	104.56	35.89	81.87
MuLim (kNm)	102	102	102	102	102	102
R	1.26	1.081	1.379	3.321	1.14	2.6
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.281	0.239	0.309	0.85	0.252	0.632
Pcclc (%) (C)	0	0	0	0.02	0	0
PtPrv (%)	0.399	0.399	0.399	1.107	1.107	1.107
AstCalc (sqmm)	238.82	203.03	263.24	723.7	214.69	537.46
AstPrv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement Provided	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement

B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	71.23	44.24	63.48
Tu (kNm)	0.25	0.25	0.63
V_{Tu} (kN)	1.72	1.72	4.4
Vut (kN)	72.95	45.96	67.88
Tv (N/sqmm)	0.86	0.54	0.8
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	16.26	0	11.19
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	238.5	153.081	234.278
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm

Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm
Side Face Reinforcement not required.		

Group	:	**G6**	
Beam No	:	**B18**	
Analysis Reference (Member)	:	96 (5.029 m)	
Beam Length	:	5067	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	13	18	15
Critical L/C - RCDC	16	9	14	9	14	11
Mu (kNm)	26.1	32.94	31.69	93.08	22.98	80.61
Tu (kNm)	0.19	0.2	0.19	0.2	0.19	0.19
M_{Tu} (kNm)	0.32	0.33	0.33	0.33	0.33	0.32
Mud (kNm)	26.43	33.27	32.02	93.41	23.31	80.94
MuLim (kNm)	102	102	102	102	102	102
R	0.839	1.057	1.017	2.967	0.74	2.57
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.233	0.224	0.742	0.2	0.623
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.399	0.399	0.399	1.107	1.107	1.107
AstCalc (sqmm)	170.2	198.12	190.24	631.08	170.2	530.17
AstPrv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement Provided	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	67.39	36.97	62.32
Tu (kNm)	0.2	0.2	0.19
V_{Tu} (kN)	1.36	1.36	1.32
Vut (kN)	68.75	38.33	63.65
Tv (N/sqmm)	0.81	0.45	0.75
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	12.06	0	6.96
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	223.628	127.35	207.319
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm
Side Face Reinforcement not required.		

Group	:	**G6**	
Beam No	:	**B19**	
Analysis Reference (Member)	:	84 (5.029 m)	
Beam Length	:	4991	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	13	18	15
Critical L/C - RCDC	16	9	14	9	14	11
Mu (kNm)	27.78	42.39	43.62	99.81	24.81	83.22
Tu (kNm)	0	0.66	0.49	0.66	0.49	0.17
M_{Tu} (kNm)	0	1.11	0.83	1.11	0.83	0.28
Mud (kNm)	27.78	43.5	44.46	100.92	25.64	83.5
MuLim (kNm)	102	102	102	102	102	102
R	0.882	1.381	1.412	3.205	0.814	2.652
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.31	0.317	0.818	0.2	0.647

Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.399	0.399	0.399	1.107	1.107	1.107
AstCalc (sqmm)	170.2	263.75	270.04	695.78	170.2	550.29
AstPrv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement Provided	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	71.7	41.74	63.02
Tu (kNm)	0.66	0.66	0.17
V_{Tu} (kN)	4.57	4.57	1.16
Vut (kN)	76.27	46.32	64.19
Tv (N/sqmm)	0.9	0.54	0.75
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	19.58	0	7.5
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	261.648	166.818	208.324
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G7**	
Beam No	:	**B20**	
Analysis Reference (Member)	:	124 (5.029 m)	
Beam Length	:	2134	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	14	12	12	12	5	5
Critical L/C - RCDC	10	8	8	8	1	1
Mu (kNm)	3.46	-	-	12.88	69.42	97.39
Tu (kNm)	6.19	6.91	6.91	6.91	0.19	0.19

M_{Tu} (kNm)	10.44	11.66	11.66	11.66	0.32	0.32
Mud (kNm)	13.9	11.66	11.66	24.54	69.74	97.7
MuLim (kNm)	102	102	102	102	102	102
R	0.442	0.37	0.37	0.779	2.215	3.103
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.523	0.785
Pcclc (%) (C)	0.047	0.079	0.079	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	170.2	445.44	667.66
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	5	5	5
Critical L/C - RCDC	1	1	1
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	44.61	50.86	52.86
Tu (kNm)	0.19	0.19	0.19
V_{Tu} (kN)	1.3	1.3	1.3
Vut (kN)	45.91	52.16	54.16
Tv (N/sqmm)	0.54	0.61	0.64
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	151.083	170.855	177.192
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81

SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G7**	
Beam No	:	**B21**	
Analysis Reference (Member)	:	109 (5.029 m)	
Beam Length	:	4496	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	13	13	15
Critical L/C - RCDC	16	9	14	9	9	11

Mu (kNm)	64.8	42.04	72.28	108.45	46.92	83.57
Tu (kNm)	0.2	0.2	0.2	0.2	0.2	0.2
M_{Tu} (kNm)	0.34	0.34	0.34	0.34	0.34	0.34
Mud (kNm)	65.14	42.38	72.62	108.79	47.26	83.91
MuLim (kNm)	102	102	102	102	102	102
R	2.069	1.346	2.306	3.455	1.501	2.665
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.484	0.301	0.549	0.884	0.339	0.65
Pcclc (%) (C)	0	0	0	0.059	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	412.05	256.45	466.79	752.26	288.57	553.57
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	8	8	10
Critical L/C - RCDC	4	4	6
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	65.06	47.77	53.29
Tu (kNm)	0.16	0.16	0.17
V_{Tu} (kN)	1.08	1.08	1.17
Vut (kN)	66.14	48.85	54.46
Tv (N/sqmm)	0.78	0.57	0.64
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	9.45	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	214.145	159.404	177.556

Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G7**	
Beam No	:	**B22**	
Analysis Reference (Member)	:	97 (5.029 m)	
Beam Length	:	5067	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design						
	Beam Bottom			Beam Top		
	Left	Mid	Right	Left	Mid	Right
Critical L/C - Analysis	20	13	18	8	13	15
Critical L/C - RCDC	16	9	14	4	9	11
Mu (kNm)	46.6	34.83	52.18	90.81	33.81	70.98
Tu (kNm)	0.1	0	0	0	0	0.12
M_{Tu} (kNm)	0.18	0	0	0	0	0.21
Mud (kNm)	46.77	34.83	52.18	90.81	33.81	71.18
MuLim (kNm)	102	102	102	102	102	102
R	1.485	1.106	1.657	2.884	1.074	2.261
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.335	0.244	0.378	0.716	0.237	0.536
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	285.33	207.98	321.62	609.39	201.53	456.11
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	Mid	Right
Critical L/C - Analysis	8	8	10
Critical L/C - RCDC	4	4	6
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	56	36.76	48.29
Tu (kNm)	0	0	0
V_{Tu} (kN)	0	0	0
Vut (kN)	56	36.76	48.29
Tv (N/sqmm)	0.66	0.43	0.57

Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	-	-	-
Asv Torsion Prv (sqmm)	-	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G7**	
Beam No	:	**B23**	
Analysis Reference (Member)	:	85 (5.029 m)	
Beam Length	:	4991	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm

Ptnominal (%) (Bn) : 0.2 %

Flexure Design						
	Beam Bottom			Beam Top		
	Left	Mid	Right	Left	Mid	Right
Critical L/C - Analysis	20	13	18	8	13	15
Critical L/C - RCDC	16	9	14	4	9	11
Mu (kNm)	43.94	44.03	63.19	89.81	31.03	73.78
Tu (kNm)	0	0.2	0.17	0.18	0.2	0
M_{Tu} (kNm)	0	0.34	0.29	0.3	0.34	0
Mud (kNm)	43.94	44.37	63.47	90.11	31.37	73.78
MuLim (kNm)	102	102	102	102	102	102
R	1.396	1.409	2.016	2.862	0.996	2.343
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.313	0.317	0.47	0.709	0.219	0.559
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	266.67	269.47	400.12	603.66	186.18	475.45
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	Mid	Right
Critical L/C - Analysis	8	8	10
Critical L/C - RCDC	4	4	6
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	58	38.94	47.6
Tu (kNm)	0.18	0.18	0
V_{Tu} (kN)	1.24	1.24	0

Vut (kN)	59.24	40.18	47.6
Tv (N/sqmm)	0.7	0.47	0.56
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	2.55	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	192.964	132.63	-
Asv Torsion Prv (sqmm)	718.14	365.6	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G8**	
Beam No	:	**B24**	
Analysis Reference (Member)	:	125 (5.029 m)	
Beam Length	:	2134	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	

Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	12	12	12	12	7	7
Critical L/C - RCDC	8	8	8	8	3	3
Mu (kNm)	10.41	-	-	9.09	54.1	78.4
Tu (kNm)	8.83	8.83	8.83	8.83	8.38	8.38
M_{Tu} (kNm)	14.91	14.91	14.91	14.91	14.15	14.15
Mud (kNm)	25.32	14.91	14.91	24	68.24	92.55
MuLim (kNm)	102	102	102	102	102	102
R	0.804	0.474	0.474	0.762	2.167	2.939
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.511	0.733
Pcclc (%) (C)	0.03	0.101	0.101	0.039	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	170.2	434.51	623.85
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**

Critical L/C - Analysis	7	7	7
Critical L/C - RCDC	3	3	3
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	39.11	43.95	45.96
Tu (kNm)	8.38	8.38	8.38
V_{Tu} (kN)	58.3	58.3	58.3
Vut (kN)	97.41	102.26	104.27
Tv (N/sqmm)	1.14	1.2	1.23
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	40.72	45.57	47.58
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	565.961	581.287	587.662
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	304.88	341.12	356.2
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G8**
Beam No	:	**B25**
Analysis Reference (Member)	:	110 (5.029 m)

Beam Length	:	4496	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	13	13	15
Critical L/C - RCDC	16	9	14	9	9	11
Mu (kNm)	53.55	41.25	58.88	120.48	43.82	95.68
Tu (kNm)	0.67	0.22	0	0.22	0.22	0.89
M_{Tu} (kNm)	1.13	0.37	0	0.37	0.37	1.5
Mud (kNm)	54.68	41.62	58.88	120.85	44.2	97.19
MuLim (kNm)	102	102	102	102	102	102
R	1.736	1.322	1.87	3.838	1.404	3.087
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.398	0.296	0.432	0.979	0.315	0.779
Pcclc (%) (C)	0	0	0	0.173	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	338.62	251.55	367.71	833.53	268.34	663.23
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement

C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	78.34	51.35	69.96
Tu (kNm)	0.22	0.22	0.89
V_{Tu} (kN)	1.53	1.53	6.19
Vut (kN)	79.86	52.88	76.15
Tv (N/sqmm)	0.94	0.62	0.89
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	23.17	0	19.46
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	259.555	174.136	268.416
Asv Torsion Prv (sqmm)	718.14	365.6	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	140
SvPrv (mm)	140	275	140
Asv Total Prv (sqmm)	718.14	365.6	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G8**	
Beam No	:	**B26**	
Analysis Reference (Member)	:	98 (5.029 m)	
Beam Length	:	5067	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	13	18	15
Critical L/C - RCDC	16	9	14	9	14	11
Mu (kNm)	36.39	40.96	45.29	109.02	30.67	93.81
Tu (kNm)	0.21	0.65	0.54	0.65	0.54	0
M_{Tu} (kNm)	0.35	1.1	0.91	1.1	0.91	0
Mud (kNm)	36.74	42.05	46.2	110.12	31.58	93.81
MuLim (kNm)	102	102	102	102	102	102
R	1.167	1.336	1.467	3.497	1.003	2.979
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.259	0.299	0.331	0.895	0.22	0.746
Pcclc (%) (C)	0	0	0	0.072	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	220.09	254.32	281.57	761.22	187.5	634.44
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	73.69	43.28	67.54
Tu (kNm)	0.65	0.65	0
V_{Tu} (kN)	4.52	4.52	0
Vut (kN)	78.21	47.79	67.54
Tv (N/sqmm)	0.92	0.56	0.79
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	21.52	0	10.85
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	267.532	171.255	-
Asv Torsion Prv (sqmm)	718.14	365.6	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm
Side Face Reinforcement not required.		

Group	:	**G8**	
Beam No	:	**B27**	
Analysis Reference (Member)	:	86 (5.029 m)	
Beam Length	:	4991	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	13	13	15
Critical L/C - RCDC	16	9	14	9	9	11
Mu (kNm)	40.78	51.24	60.25	114.7	31.36	99.27
Tu (kNm)	0.66	1.04	0.61	1.04	1.04	1.09
M_{Tu} (kNm)	1.12	1.75	1.03	1.75	1.75	1.84
Mud (kNm)	41.9	52.99	61.28	116.45	33.11	101.11
MuLim (kNm)	102	102	102	102	102	102
R	1.331	1.683	1.946	3.698	1.052	3.211
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.298	0.384	0.452	0.945	0.232	0.82
Pcclc (%) (C)	0	0	0	0.132	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	253.35	327.08	384.56	803.91	197.13	697.5

AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	78.04	48.08	69.3
Tu (kNm)	1.04	1.04	1.09
V_{Tu} (kN)	7.22	7.22	7.57
Vut (kN)	85.27	55.31	76.87
Tv (N/sqmm)	1	0.65	0.9
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	28.58	0	20.18
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	301.816	206.987	276.758
Asv Torsion Prv (sqmm)	718.14	365.6	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	140
SvPrv (mm)	140	275	140
Asv Total Prv (sqmm)	718.14	365.6	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G9**	
Beam No	:	**B28**	
Analysis Reference (Member)	:	169 (5.029 m)	
Beam Length	:	2134	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	7	7	7	8	8	8
Critical L/C - RCDC	3	3	3	4	4	4
Mu (kNm)	-	-	-	11.62	27.1	38.56
Tu (kNm)	9.1	9.1	9.1	8.65	8.65	8.65
M_{Tu} (kNm)	15.37	15.37	15.37	14.6	14.6	14.6
Mud (kNm)	15.37	15.37	15.37	26.21	41.7	53.16
MuLim (kNm)	102	102	102	102	102	102

R	0.488	0.488	0.488	0.833	1.324	1.688
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.296	0.386
Pcclc (%) (C)	0.104	0.104	0.104	0.02	0	0
PtPrv (%)	0.266	0.266	0.266	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	170.2	252.06	328.26
AstPrv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement Provided	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	13	8	8
Critical L/C - RCDC	9	4	4
PtPrv (%)	0.399	0.399	0.399
Vu (kN)	11.9	17.83	23.31
Tu (kNm)	8.12	8.65	8.65
V_{Tu} (kN)	56.48	60.16	60.16
Vut (kN)	68.38	77.99	83.47
Tv (N/sqmm)	0.8	0.92	0.98
Tc (N/sqmm)	0.45	0.45	0.45
Vc (kN)	37.88	37.88	37.88
Vus=Vut-Vc (kN)	30.5	40.11	45.59
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	434.353	478.64	495.608
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	300.21	341.25
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G9**	
Beam No	:	**B29**	
Analysis Reference (Member)	:	170 (5.029 m)	
Beam Length	:	4496	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	20	12	13	13	15

Critical L/C - RCDC	16	16	8	9	9	11
Mu (kNm)	10.93	10.93	0.42	45.3	20.82	19.16
Tu (kNm)	1.43	1.43	1.69	0.18	0.18	1.75
M_{Tu} (kNm)	2.42	2.42	2.85	0.3	0.3	2.95
Mud (kNm)	13.35	13.35	3.27	45.61	21.12	22.11
MuLim (kNm)	102	102	102	102	102	102
R	0.424	0.424	0.104	1.448	0.671	0.702
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.326	0.2	0.2
Pcclc (%) (C)	0	0	0.016	0	0	0
PtPrv (%)	0.266	0.266	0.266	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	277.63	170.2	170.2
AstPrv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement Provided	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	5	13	15
Critical L/C - RCDC	1	9	11
PtPrv (%)	0.399	0.399	0.399
Vu (kN)	26.68	14.77	20.84
Tu (kNm)	0.9	0.18	1.75
V_{Tu} (kN)	6.24	1.25	12.16
Vut (kN)	32.92	16.02	33.01
Tv (N/sqmm)	0.39	0.19	0.39
Tc (N/sqmm)	0.45	0.45	0.45
Vc (kN)	37.88	37.88	37.88
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8

Asv Torsion (sqmm)	126.464	54.503	150.121
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G9**	
Beam No	:	**B30**	
Analysis Reference (Member)	:	171 (5.029 m)	
Beam Length	:	5067	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design						
	Beam Bottom			**Beam Top**		

	Left	Mid	Right	Left	Mid	Right
Critical L/C - Analysis	5	5	13	8	13	5
Critical L/C - RCDC	1	1	9	4	9	1
Mu (kNm)	5.01	13.44	6.54	20.15	-	20.42
Tu (kNm)	0.71	0.71	1.06	1	1.06	0.71
M_{Tu} (kNm)	1.19	1.19	1.78	1.68	1.78	1.19
Mud (kNm)	6.2	14.63	8.33	21.83	1.78	21.62
MuLim (kNm)	102	102	102	102	102	102
R	0.197	0.465	0.264	0.693	0.057	0.686
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2
Pcclc (%) (C)	0	0	0	0	0.012	0
PtPrv (%)	0.266	0.266	0.266	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
AstPrv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement Provided	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	5	8	5
Critical L/C - RCDC	1	4	1
PtPrv (%)	0.399	0.266	0.399
Vu (kN)	25.73	8.82	25.82
Tu (kNm)	0.71	1	0.71
V_{Tu} (kN)	4.91	6.94	4.91
Vut (kN)	30.64	15.76	30.73
Tv (N/sqmm)	0.36	0.19	0.36
Tc (N/sqmm)	0.45	0.37	0.45
Vc (kN)	37.88	31.88	37.88
Vus=Vut-Vc (kN)	0	0	0

Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	114.176	76.127	114.475
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G9**	
Beam No	:	**B31**	
Analysis Reference (Member)	:	172 (5.029 m)	
Beam Length	:	4991	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	5	5	8	8	5	15
Critical L/C - RCDC	1	1	4	4	1	11
Mu (kNm)	6.46	19.95	18.75	23.29	-	15.62
Tu (kNm)	2.62	2.62	2.28	2.28	2.62	1.72
M_{Tu} (kNm)	4.43	4.43	3.84	3.84	4.43	2.91
Mud (kNm)	10.89	24.38	22.59	27.13	4.43	18.53
MuLim (kNm)	102	102	102	102	102	102
R	0.346	0.774	0.718	0.862	0.141	0.589
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2
Pcclc (%) (C)	0	0	0	0	0.03	0
PtPrv (%)	0.266	0.266	0.266	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
AstPrv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement Provided	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	5	8	5
Critical L/C - RCDC	1	4	1
PtPrv (%)	0.399	0.266	0.399
Vu (kN)	29.08	13.05	21.67
Tu (kNm)	2.62	2.28	2.62
V_{Tu} (kN)	18.25	15.84	18.25
Vut (kN)	47.33	28.89	39.92
Tv (N/sqmm)	0.56	0.34	0.47
Tc (N/sqmm)	0.45	0.37	0.45

Vc (kN)	37.88	31.88	37.88
Vus=Vut-Vc (kN)	9.45	0	2.04
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	218.451	151.862	195.528
Asv Torsion Prv (sqmm)	718.14	365.6	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	140
SvPrv (mm)	140	275	140
Asv Total Prv (sqmm)	718.14	365.6	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.